



Transcriptomics and systems network-based molecular mechanism of herbal formula Huosu-Yangwei inhibited gastric cancer *in vivo*

Mengyuan Zhang^{a,1}, Yujie Ding^{b,1}, Sheng Hu^c, Fulong Li^a, Yi Wang^a, Yue Zhou^a, Mei Qi^a, HongMei Ni^{b,***}, Shengquan Fang^{a,**}, Qilong Chen^{c,d,*}

^a Department of Gastroenterology, Yueyang Hospital of Integrated Traditional Chinese and Western Medicine, Shanghai University of Traditional Chinese Medicine, Shanghai, 200437, China

^b School of Basic Medical Science, Shanghai University of Traditional Chinese Medicine, Shanghai, 201203, China

^c Central Laboratory, Shanghai Skin Disease Hospital, School of Medicine, Tongji University, Shanghai, 200443, China

^d Shanghai Institute of Stem Cell Research and Clinical Translation, Shanghai, 200120, China

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ABSTRACT

Ethnopharmacological relevance: The efficacy of the herbal formula Huosu-Yangwei (HSYW) in the treatment of advanced gastric cancer and chronic atrophic gastritis with precancerous lesions has been reported in clinical trials. However, the molecular mechanisms underlying its inhibition of gastric tumor are not well-understood.

Aim of the study: Combined with transcriptomics and systems network-based molecular mechanism to explore the potential circRNA-miRNA-mRNA network of HSYW in the treatment of gastric cancer.

Materials and methods: Animal experiments were conducted to investigate the effect of HSYW on tumor growth *in vivo*. RNA sequencing (RNA-seq) was implemented to identify the differentially expressed (DE) genes. Predictive miRNA targets and mRNA were used to construct circRNA-miRNA-mRNA networks and protein-protein interaction (PPI) networks. Quantitative real-time PCR (qRT-PCR) was utilized to verify the accuracy of the proposed circRNA-miRNA-mRNA networks. Additionally, the differentially expressed target proteins between gastric cancer (GC) and normal patients were assessed using data from the TCGA (The Cancer Genome Atlas) and HPA (The Human Protein Atlas) databases.

Results: We demonstrate HSYW significantly inhibits tumor growth of N87 cell-bearing Balb/c mice. Transcriptomic analysis revealed the existence of 119 differentially expressed (DE) circRNAs and 200 DE mRNAs between HSYW-treated and model mice. By associating predicted circRNA-miRNA pairs and miRNA-mRNA pairs, we constructed a circRNA-miRNA-mRNA (CMM) network. Furthermore, a protein-protein interaction (PPI) network was developed using the differential expressed mRNAs. Consequently, the reconstructed core CMM network and qRT-PCR validation indicated that 4 circRNAs, 5 miRNAs and 6 mRNAs could potentially serve as biomarkers to assess the therapeutic effects of HSYW-treated N87-bearing Balb/c mice. The TCGA and HPA databases also demonstrated that mRNA KLF15 and PREX1 had substantial differences between gastric cancer (GC) and healthy controls.

Conclusions: By combining the experimental and bioinformatics analysis, this study confirms that the circRNA_00240/hsa-miR-642a-5p/KLF15 and circRNA_07980/hsa-miR-766-3p/PREX1 pathways play critical roles in HSYW-treated gastric cancer.

Abbreviations: ceRNAs, Competing for endogenous RNAs; CMM, circRNA-miRNA-mRNA; CYPs, Cytochrome P450 enzymes; DE, Differentially expressed; EMT, Epithelial-mesenchymal transition; FPKM, Fragments Per kb Per Million Reads; GC, Gastric cancer; GEFs, Guanine nucleotide exchange factors; GEPIA, Gene Expression Profiling Interactive Analysis; GTEx, Genotype-Tissue Expression; GO, Gene Ontology; HPA, The Human Protein Atlas; HPLC, High Performance Liquid Chromatography; HSYW, Huosu-Yangwei; KEGG, Kyoto Encyclopedia of Genes and Genomes; MF, Molecule function; NB, Negative binomial distribution; OXA, Oxaliplatin; PPI, Protein-protein interaction; RNA-seq, RNA sequencing; RPM, Spliced reads per million; SOX, S-1 plus oxaliplatin; TCGA, The Cancer Genome Atlas; TCM, Traditional Chinese medicine.

* Corresponding author. Central Laboratory, Shanghai Skin Disease Hospital, School of Medicine, Tongji University, Shanghai, 200443, China.

** Corresponding author.

*** Corresponding author.

E-mail addresses: nihm20032003@163.com (H. Ni), fsq20032003@163.com (S. Fang), cqlw1975@126.com (Q. Chen).

¹ These authors contributed equally to this work and should be considered co-first authors.

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1. Introduction

Gastric cancer (GC) is a prevalent malignancy globally, accounting for the fifth highest rate of new diagnoses and the third highest rate of cancer-related mortality (Bray et al., 2018). According to the Global Burden of Disease Study (GBD 2019 Diseases and Injuries Collaborators, 2020) (Global burden of 369 diseases and injuries in 204 countries and territories, 2020), 44.21% of Chinese Disability-Adjusted Life Years (DALYs) are attributed to GC, and approximately 45% of Chinese cancer fatalities were caused by gastrointestinal cancers (Cao et al., 2021). It is estimated that by 2022, there will be over 509,421 deaths due to GC in China (Xia et al., 2022). Chemotherapy is the mainstay of GC treatment, with platinum-based drugs being widely utilized as first-line combined regimens (Wang et al., 2021). For example, Oxaliplatin is commonly used in many current chemotherapy regimens (Wagner et al., 2017; Joshi and Badgwell, 2021). A clinical trial showed that the overall survival rate of Oxaliplatin-based regimens was 16.0 months, compared to the 15.1 months seen in patients treated with S-1 plus cisplatin (Kang et al., 2020). However, chemotherapy causes cancer patients to experience many specific side effects (Oun et al., 2018), such as nephrotoxicity (Crona et al., 2017), myelosuppression (Lv et al., 2018), neurotoxicity (Avan et al., 2015; Kanat et al., 2017), which necessitates the urgent development of novel drugs or strategies for GC treatment.

Herbal formula has advantages in preventing tumor recurrence and metastasis, lessening adverse reactions, and prolonging the overall survival of cancer patients (Jiao et al., 2018; Wong et al., 2019; Zhang et al., 2020; Tang et al., 2020; Zhu et al., 2021b). Huosu-Yangwei (HSYW) formula, has been widely used in clinical practice for more than two decades as a treatment for atrophic gastritis with precancerous lesions (Guo et al., 2015; Ye et al., 2015) and gastric cancer (Zhu et al., 2021a). Additionally, a combination of HSYW with S-1 plus oxaliplatin (SOX) has been demonstrated to diminish clinical symptoms, as well as enhance survival quality and reduce the incidence of adverse medication responses (anemia, nausea, and vomiting) in gastric cancer patients. We further confirmed HSYW could inhibit cytochrome P450 enzymes (CYPs) and modulated the metabolic half-life of apatinib (Fang et al., 2020). Network pharmacology analysis identify that the HSYW formula may significantly inhibit the growth of gastric cancer via multi-target and multi-pathway (Fang et al., 2021; Zhao et al., 2021). However, the molecular mechanisms of HSYW-inhibited tumor growth of gastric cancer remain unclear.

CircRNAs are an abundant class of endogenous RNAs that play a critical role in various cancers as biomarkers and in carcinogenesis through multiple distinct mechanisms (Liu et al., 2021). CircRNAs are typically generated from the splicing of exons in a head-to-tail configuration and function as transcriptional regulators, miRNA sponges, and bind to proteins in several ways. These RNAs are crucial for tumor proliferation, apoptosis, migration, and invasion (Li et al., 2020a; Wang et al., 2022). Moreover, circRNAs are known to bind miRNAs and act as competing endogenous RNAs (ceRNAs) which can reduce the repressive effect of miRNAs on target genes, thereby raising expression levels (Li et al., 2020b). The ceRNA regulatory models have identified circRNA as miRNA sponges that indirectly regulate miRNA downstream target genes (Zeng et al., 2022). In gastric cancer, circAKT3 has been found to sponge miR-198 and activate PIK3R1 expression to provide resistance to cisplatin and repress tumor apoptosis (Huang et al., 2019). Additionally, circNRIP1 works as a tumor promoter by sponging miR-149-5p to alter AKT1 expression levels to promote gastric cancer progression (Zhang et al., 2019). Moreover, the herbal formula exhibited substantial anti-hepatocellular carcinoma effects through enhancing the lipid and glucose metabolism of liver via circRNA-miRNA-mRNA networks (Liu et al., 2022). It is suggested that the circRNAs can act as biomarkers and play critical roles for GC treatment.

In this work, we focus on the expression levels of circRNAs in HSYW-treated gastric cancer. Transcriptomics and systems biology were used to construct potentially CMM (circRNA-miRNA-mRNA, CMM) network

and PPI (protein-protein interaction, PPI) network, and further identify several biomarkers during the efficacy assessment of HSYW-treated gastric cancer. This study can help us to reveal new targets and strategies for gastric cancer treatment.

2. Materials and methods

2.1. Animal

Male Balb/c mice (4 weeks old) were purchased from Shanghai Model Organisms Center (SCXK Shanghai, 2014-0002) and fed in the SPF Animal Laboratory of Shanghai University traditional Chinese medicine, (SYXK Shanghai, 2014-0008). In a normal air-conditioned environment (23 ± 2 °C, 55 ± 10% R.H.), the mice were kept in polycarbonate cages. The Shanghai University of Traditional Chinese Medicine Ethical Committee on Animal Experiments, located in Shanghai, China, approved all animal treatments. The study follows the 3Rs approach and the ARRIVE criteria for animal experimentation.

2.2. Mice model

Our previous studies demonstrated that the HSYW was significantly inhibited N87 cell growth compare with SGC-7901, HGC27, AGS and BGC823 cells (Zhao et al., 2021). The results suggested that N87 cell was the best suitable model *in vitro*. In this work, the Institute of Biochemistry and Cell Biology provided the human gastric cancer cell line N87 (Shanghai, China). At 37 °C in a humidified 5% CO₂ incubator, the cell line N87 was grown in DMEM medium (Gibco, USA) supplemented with 10% fetal bovine serum (FBS) (Gibco, USA) and 100U/mL penicillin-streptomycin (Beyotime, China). The mice were weighed and randomly divided into 5 groups, including Oxaliplatin (OXA) group (n = 10), model group (n = 10), HSYW low dose group (n = 10), HSYW medium dose group (n = 10), and HSYW high dose group (n = 10). The cells at the logarithmic phase (5 × 10⁶/mL) were subcutaneously inoculated in the right axillary of the mice (100 µL per mouse). Then, all animals were anesthetized under pentobarbital intraperitoneal injection (150 mg/kg), and blood samples and gastric tumor tissues were collected and stored at -80 °C for the following analyses.

2.3. Huosu-yangwei formula preparation

Huosu-Yangwei (HSYW) formula consists of 21 kinds of herbs, as shown in Table 1. Here, the HSYW preparation was produced by Shanghai Baolong Pharmaceutical Co., Ltd. (trade name: Huosu Yangwei Oral Solution, batch number: Z05050328). In previous work, we constructed the quantitative analysis of HSYW formula, and 160 ingredients of HSYW were identified using the HPLC/MS (Fang et al., 2021). The linear range of astragaloside in HSYW was 0.262–15.696 µg with average recovery of 99.46% (Ding et al., 2015).

2.4. Drug treatment

The mice were administered the same dose of HSYW as the clinical equivalent of humans (body weight, 60 kg), with a conversion coefficient of 10. Accordingly, the total dose of 20 g mouse was 0.2 mL (60 mL/60 kg × 0.02 kg × 10.0) daily. The model group mice were given 0.1% sodium chloride solution (0.2 mL per mouse, once daily) for 25 days. The mice in the oxaliplatin group were treated with 6 mg/(kg d) oxaliplatin in 0.1 mL saline and injected intraperitoneally on the 2nd day after inoculation, and then injected every 12 days. One week post-inoculation, tumor were measured with a vernier caliper every three day intervals, and mice weighed every six days. The tumor volume was calculated as follows: tumor volume = $a \times b^2/2$. (a represents the longest diameter, and b represents the shortest diameter of the tumor). Finally, the mice were euthanized 24 days after injection, the tumors were removed and weighed, and tumor volume was evaluated.

Table 1
Plant name cross-reference table.

Chinese name	Latin name
Huoxiang	<i>Pogostemon cablin</i> (Blanco) Benth.
Zisugeng	<i>Perilla frutescens</i> (L.) Britton.
Baizhu	<i>Atractylodes macrocephala</i> Koidz.
Zhike	<i>Citrus trifoliata</i> L.
Doukou	<i>Amomum kwangsiense</i> D.Fang & X.X.Chen.
Foshou	<i>Citrus medica</i> L.
WuMei	<i>Prunus mume</i> (Siebold) Siebold & Zucc.
Shengjiang	<i>Zingiber officinale</i> Roscoe.
Dazao	<i>Ziziphus jujuba</i> Mill.
Gancao	<i>Glycyrrhiza uralensis</i> Fisch. ex DC.
Huangqi	<i>Astragalus mongholicus</i> Bunge.
Dihuang	<i>Rehmannia glutinosa</i> (Gaertn.) DC.
Mudanpi	<i>Paeonia × suffruticosa</i> Andrews.
Tianhuafen	<i>Trichosanthes kirilowii</i> Maxim.
Danggui	<i>Angelica sinensis</i> (Oliv.) Diels.
Chuanxiong	<i>Conioselinum anthriscoides</i> ‘Chuanxiong’ S. H.
Ezhu	<i>Curcuma zedoaria</i> (Christm.) Roscoe.
Gouqizi	<i>Lycium chinense</i> Mill.
Huanglian	<i>Coptis chinensis</i> Franch.
Dangsheng	<i>Codonopsis pilosula</i> (Franch.) Nannf.
Pugongying	<i>Taraxacum mongolicum</i> Hand.-Mazz.

2.5. Transcriptional profiles detection

Using RNA sequencing, the transcriptional profiles of HSYW-treated gastric cancer mice (the HSYW Group, HSYW, n = 3) and gastric cancer mice (the Model Group, M, n = 3) were detected, adopting TRIzol reagent (Invitrogen, Grand Island, NY, USA), respectively. Following the manufacturer’s instructions, total RNA was isolated from the tumor tissues using the total RNA Isolation Kit (Ambion). RNA integrity was evaluated using the Agilent 2100 Bioanalyzer (Agilent Technologies, Santa Clara, CA, USA). The samples with RNA Integrity Number (RIN) ≥ 7 were subjected to subsequent analysis. The libraries were constructed using TruSeq Stranded and then sequenced on the Illumina Sequencing Platform (HiSeq™ 2500). Raw reads generated during high throughput sequencing were fastq format sequences. The raw reads required additional filtering to provide high-quality reads for subsequent analysis. The sequencing data statistics are shown in supplementary attachment (S1). The RNA-Seq data were deposited in Genome Sequence Archive (GSA) database, and the accession number is [CRA010889](https://bigd.big.ac.cn/project/GSA010889).

2.6. Differentially expressed circRNAs and mRNAs analysis

The circRNA expression levels were determined using the RPM (spliced reads per million) algorithm, CIRI program and circBase database. The mRNA expressions were calculated with FPKM (Fragments Per kb Per Million Reads) algorithm. The circRNA and mRNA counts of all samples were normalized, and the difference significance test of reads was carried out using NB (Negative binomial distribution), respectively. The differentially expressed (DE) circRNAs and mRNAs between the HSYW group and model group ($P < 0.05$, Fold-Change >1.5) were calculated by R package. Using the programs Cluster 3.0 and Java TreeView, heat maps and hierarchical analyses were conducted. Afterward, the biological functions of DE circRNAs and mRNAs were analyzed with DAVID online (<https://david.ncifcrf.gov/>) ($P < 0.05$), respectively.

2.7. circRNA-miRNA-mRNA network construction

Using Miranda software and TarBase (v8.0) database, the circRNA-miRNA pairs and miRNA-mRNA pairs were predicted, respectively. Furthermore, a CMM (circRNA-miRNA-mRNA, CMM) network was constructed based on the predicted circRNA-miRNA pairs and miRNA-mRNA pairs. Furthermore, the network parameters were calculated, which including betweenness centrality (BC), closeness centrality (CC),

degree (De) and topological coefficient (TC), and the core node was defined as $BC \geq \text{avg}(BC)$, $CC \geq \text{avg}(CC)$, $De \geq \text{avg}(De)$, and $TC \geq \text{avg}(TC)$. Finally, the rebuilt CMM network was considered as a core network, which might high correlate with HSYW treated gastric cancer.

2.8. Protein-protein interaction network construction and core nodes selection

In this work, the DE mRNAs related protein-protein interaction (PPI) network was constructed by IntAct (Kerrien et al., 2012), BioGrid (Oughtred et al., 2019) and MINT (Górska et al., 2015) databases. The core PPI network was calculated by ClusterONE algorithm (Nepusz et al., 2012), which were defined as $P < 0.001$, quality >0.60 , node size >5 , and network density >0.05 . Finally, the core PPI network was mapped with core ceRNA network, respectively. The obtained mRNAs might act as neo-biomarkers, and these mRNAs could be play critical roles in HSYW treated gastric cancer.

2.9. qRT-PCR validation

Using SYBR Green PCR Master Mixture (TOYOBO, LTD, Japan) according to the manufacturer’s instructions on a Rota-Gene 6000 Real-time PCR machine, the hub circRNAs and mRNAs in HSYW formula-treated GC mice were identified. Besides, the ABI kit was utilized to make Real-Time PCR amplification of the hub circRNAs, miRNAs, and mRNAs. To test the specificity of each PCR product, a melting curve was constructed, and the required number of cycles, Ct values was determined. All circRNAs, miRNAs, and mRNAs were validated in triplicate, and the mRNA levels were calculated using $2^{-\Delta\Delta Ct}$. The gene levels were estimated using the Student’s t-test, and differences with P value < 0.05 were considered significant. The primer sequences were designed in the laboratory and synthesized by Generay Biotech (Generay, PRC) based on the RNA sequences obtained from the NCBI database (Table 2). The expression levels of mRNAs were normalized to ACTB. The miRNA primer sequences were obtained from the miRBase database in supplemental attachment (S2).

2.10. Dual-luciferase reporter assay

The conjunct sites of CircRNA_07980/miR-766-3p/PREX1 and circRNA_00240/hsa-miR-642a-5p/KLF15 were predicted by Starbase (<http://starbase.sysu.edu.cn/>). The wild type and mutant plasmids of CircRNA_07980 (psiCHECK-2-WT-CircRNA_07980, psiCHECK-2-MUT-CircRNA_07980), CircRNA_00240 (psiCHECK-2-WT-CircRNA_00240, psiCHECK-2-MUT-CircRNA_00240), PREX1 (psiCHECK-2-WT-PREX1 3’UTR, psiCHECK-2-MUT-PREX1 3’UTR), and KLF15 (psiCHECK-2-WT-KLF15 3’UTR, psiCHECK-2-MUT-KLF15 3’UTR) were offered by Yilaibo (<http://www.shyilaibo.com>), respectively. Each plasmid was co-transfected with miR-NC or miR-766-3p/miR-642a-5p for 48 h, and

Table 2
The primer sequences were obtained from the NCBI database.

Name	Sense Primer	Anti-sense Primer
circRNA_00240	TCAAGTCTTCGCCAATCACA	AAATAGCCAGACAGCACCCC
circRNA_02049	CCGCACATAGAACTTCTGCA	TCAGTGCTTCCACTTCGCTT
circRNA_03225	CATGGCACTGATCCCCCTG	TAGGCCGGATGCTTGAGATG
circRNA_07464	GAGATCTCCAGCCTTACCCG	CCAGGTTGTGCAGCTTGTTT
circRNA_07489	CAGGATGGATGGTGACAT	TGGAGTGTCTAAGAAGTGTT
circRNA_07980	ATCTCCAACACCGTCCAGC	GCTTGCTGCTGCTCTTGAC
circRNA_16548	AGCCAAAGAAATGGGTTTTC	AGGCAACATTAGGCACCTTC
circRNA_17470	CAGGATTCTGTATAAAGGAG	ATGTCCCTAGGAGTGCTT
KLF15	GAGTGCTGGAGGAATGTC	GCATCGCTGTCATCTTCA
HCN2	AGCGACTTCAGGTCTACT	GCAGTGTCTCATCTTCTG
GRAMD1B	CCTGTGGAAGAGAATGAAGT	AGTGTGCTGTTGGTGATG
CSPG5	TGAAGAGGAGGAGGAAGAG	GGAAGTCGGTGATGATGG
RRM2	GCAAGCGATGGCATAGTA	TTCTATCCGAACAGCATTGA
PRES1	GGACAGCGGATTACCATAG	CCAGGCAACGAACCTATT

then the luciferase activity was evaluated using dual-luciferase reporter kit (E1901; Promega, USA).

2.11. Survival analysis for core genes expression in GC patients

The GEPIA (Gene Expression Profiling Interactive Analysis) database (<http://gepia.cancer-pku.cn/>) was used to evaluate the expression levels of core proteins of GC patients on clinical prognosis, which analysis RNA sequencing expression data of 9736 tumors and 8587 normal samples from the TCGA (The Cancer Genome Atlas) and the GTEx (Genotype-Tissue Expression) projects. The UALCAN database (<http://ualcan.path.uab.edu/analysis.html>) was used to analyze the expression levels of core proteins of GC patients in different tumor grade. Afterward, the core proteins related immunohistochemical results between gastric glandular cells and tumor cell were also used from The Human Protein Atlas (HPA) database (<https://www.proteinatlas.org>).

2.12. Statistical analysis

Statistical analysis was conducted by SPSS 23.0 software (IBM Analytics), and all values are presented as mean $\pm$ SD with three independent experiments. The two-tailed Student's *t*-tests were used to evaluate the differences between groups, and *P* value < 0.05 was considered statistically significant.

3. Results

3.1. HSYW-inhibited tumor growth in N87-bearing Balb/c mice

The N87-bearing Balb/c mice were used to evaluate the effects of HSYW on the inhibition of gastric cancer. After 25 days, the tumor volumes revealed that the HSYW group tended to inhibit the proliferation of gastric cancer cells, with the most significant efficacy being observed in the medium-dose group (Fig. 1A). In this work, the OXA group was defined as positive control, and we notice that the tumor suppressive effect of medium-dose HSYW was very close to the positive drug (Fig. 1B, *p* = 0.124; Fig. 1C, *p* = 0.054). Remarkably, the HSYW group mice exhibited stable weight changes while OXA group mice showed continuous weight loss (Fig. 1D). These results revealed that the HSYW formula not only inhibited tumor cells growth, but also effectively preserved the body weight *in vivo*.

3.2. CircRNAs profiling analysis

In this work, the transcriptional profiles between HSYW treated (*n* = 3) and N87-bearing Balb/c mice (*n* = 3) were detected by RNA sequencing. Using the R package, 119 differentially expressed (DE) circRNAs were identified (*P* < 0.05, fold change >2), including 64 upregulated and 55 downregulated circRNAs (Fig. 1E). Enrichment analysis showed that the top 10 GO terms were mainly associated with mitotic cell cycle, oxidoreductase activity, extracellular matrix organization, protein complex assembly, structural constituent of the cytoskeleton, midbody, antigen processing and presentation of exogenous peptide antigen via MHC class II, microtubule cytoskeleton, insulin receptor signaling pathway, ER to Golgi vesicle-mediated transport (Fig. 1F). The top 10 KEGG pathways are mainly related to steroid biosynthesis, mitophagy, vasopressin-regulated water reabsorption, collecting duct acid secretion, nicotinamide metabolism, other glycan degradation, neuroactive ligand-receptor interaction, rheumatoid arthritis, carbohydrate digestion and absorption, cardiac muscle contraction (Fig. 1G). The specific classification of the GO analysis was shown in the supplementary attachment (S3). such as molecular function, biological process, and cell composition. Furthermore, we noticed

that the KEGG classification of DE circRNAs was mainly associated with human diseases, cellular processes, organismal systems, in metabolism, genetic information processing, and environmental information processing (Fig. 1H).

3.3. mRNAs profiling analysis

Using the transcriptional profiles, we further obtained 200 DE mRNAs (Fold change >1.5, *P* < 0.05) from HSYW formula-treated gastric cancer mice, including 97 upregulated and 103 downregulated mRNAs (Fig. 1I). Enrichment analysis showed that the GO terms were mainly associated with 25-hydroxycholecalciferol-24-hydroxylase activity, 1- α ,25-dihydroxy vitamin D3 24-hydroxylase activity, vitamin D catabolic process, vitamin D receptor signaling pathway, regulation of mesenchymal cell proliferation, racemase and epimerase activity, microglial cell activation involved in immune response, microglial cell proliferation, positive regulation of toll-like receptor 9 signaling pathway, CD4-positive, alpha-beta T cell activation (Fig. 1J). The KEGG pathways mainly associated with cytosolic DNA-sensing pathway, influenza A, retinol metabolism, Pertussis, Salmonella infection, steroid biosynthesis, Arginine biosynthesis, ascorbate and aldarate metabolism, Phototransduction, pentose and glucuronate interconversions (Fig. 1K). And then, the KEGG classification of DE mRNAs were Metabolism, Human diseases, Organismal systems, Cellular processes, Environmental information processing (Fig. 1L).

3.4. CircRNA-miRNA-mRNA network construction

The Miranda program and TarBase (v8.0) database were used to predict circRNA-miRNA and miRNA-mRNA pairs, and then the circRNA-miRNA network (Fig. 2A) and miRNA-mRNA network (Fig. 2B) were constructed, respectively. Combining with the circRNA-miRNA network and miRNA-mRNA network, the circRNA-miRNA-mRNA (CMM) network, also known as ceRNAs network was subsequently constructed (Fig. 2C). To comprehensively investigate the essential targets, the core nodes of CMM network were defined as BC $\geq$ avg (BC), CC $\geq$ avg (CC), De $\geq$ avg (De), and TC $\geq$ avg (TC). Finally, the core CMM networks were rebuilt (Fig. 2D and E), from the original network and considered as high associated with the HSYW-treated gastric cancer.

3.5. PPI network construction and core genes selection

Based on the IntAct, BioGRID, and MINT databases, a protein-protein interaction (PPI) network of HSYW group was constructed (Fig. 2F). Using ClusterONE algorithm, the core-PPI network was further isolated (Fig. 2G), which defined as node >5, *p* < 0.001, density >0.05, quality >0.60. Afterward, the core PPI network and core CMM network were mapped, and the obtained up mapped CMM network and down mapped CMM network were considered as play critical roles in HSYW treatment process (Fig. 2H and I). In up mapped CMM network, critical genes including circRNA_00240, circRNA_03225, circRNA_02049, hsa-miR-642a-5p, hsa-miR-671-5p, hsa-miR-339-5p, KLF15, DNAJB9, HOXC6, MAFB, HCN2 and CSPG5. The down mapped CMM network related genes including circRNA_07980, circRNA_17470, circRNA_07464, circRNA_16548, circRNA_07489, hsa-miR-3158-3p, hsa-miR-361-3p, hsa-miR-671-5p, hsa-miR-766-3p, hsa-miR-76, HEY1, MRC2, PREX1, C2orf48, NRG1, GRAMD1B and RIMKLB. These genes act as potential biomarkers and might play important roles in HSYW treated process.

3.6. Quantitative real-time PCR

The quantitative real-time PCR (qRT-PCR) was performed to validate the expression levels of circRNAs, miRNAs and mRNAs from core CMM

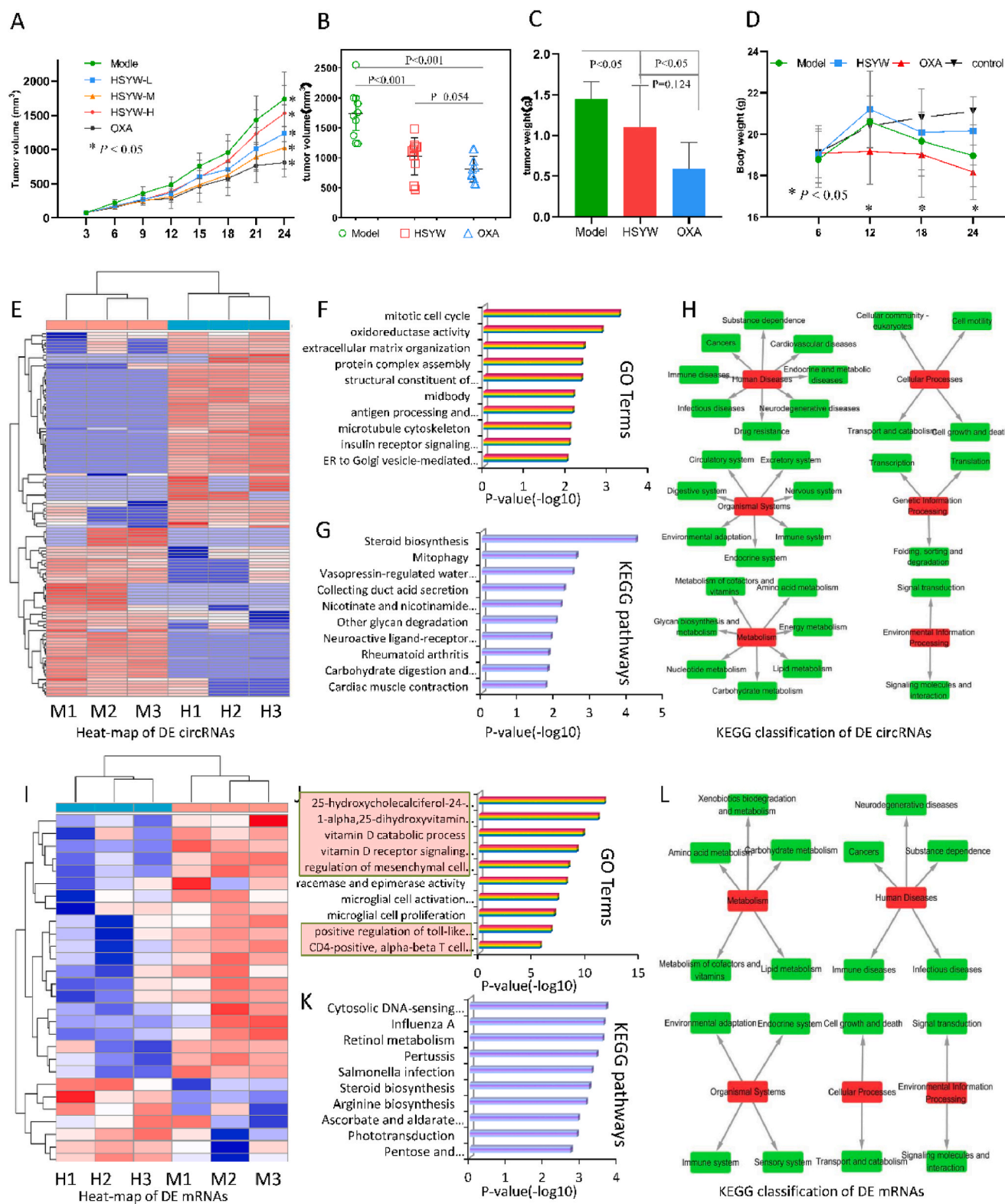


Fig. 1. HSYW inhibits tumor growth in N87-bearing Balb/c mice and circRNAs detection. (A) Growth curves of tumor volume among the low dose HSYW (HSYW-L), medium dose HSYW (HSYW-M), high dose HSYW (HSYW-H), OXA and Model groups; (B) Tumor volume among the HSYW, OXA and Model groups; (C) Tumor weight among the HSYW, OXA and Model groups; (D) Mice weight among the HSYW, OXA, Model, and control group; (E) Heat-map of differentially expressed circRNA between the HSYW and Model groups. The red color means the circRNA is up-expressed, and the blue means it is down-expression; (F) Differentially expressed circRNAs related Gene Ontology terms; (G) DE circRNAs related KEGG pathways; (H) KEGG classification of DE circRNAs; (I) Heat-map of differentially expressed mRNA between the HSYW and Model groups; (J) Differentially expressed mRNAs related Gene Ontology terms; (K) DE mRNAs related KEGG pathways; (L) KEGG classification of DE mRNAs. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

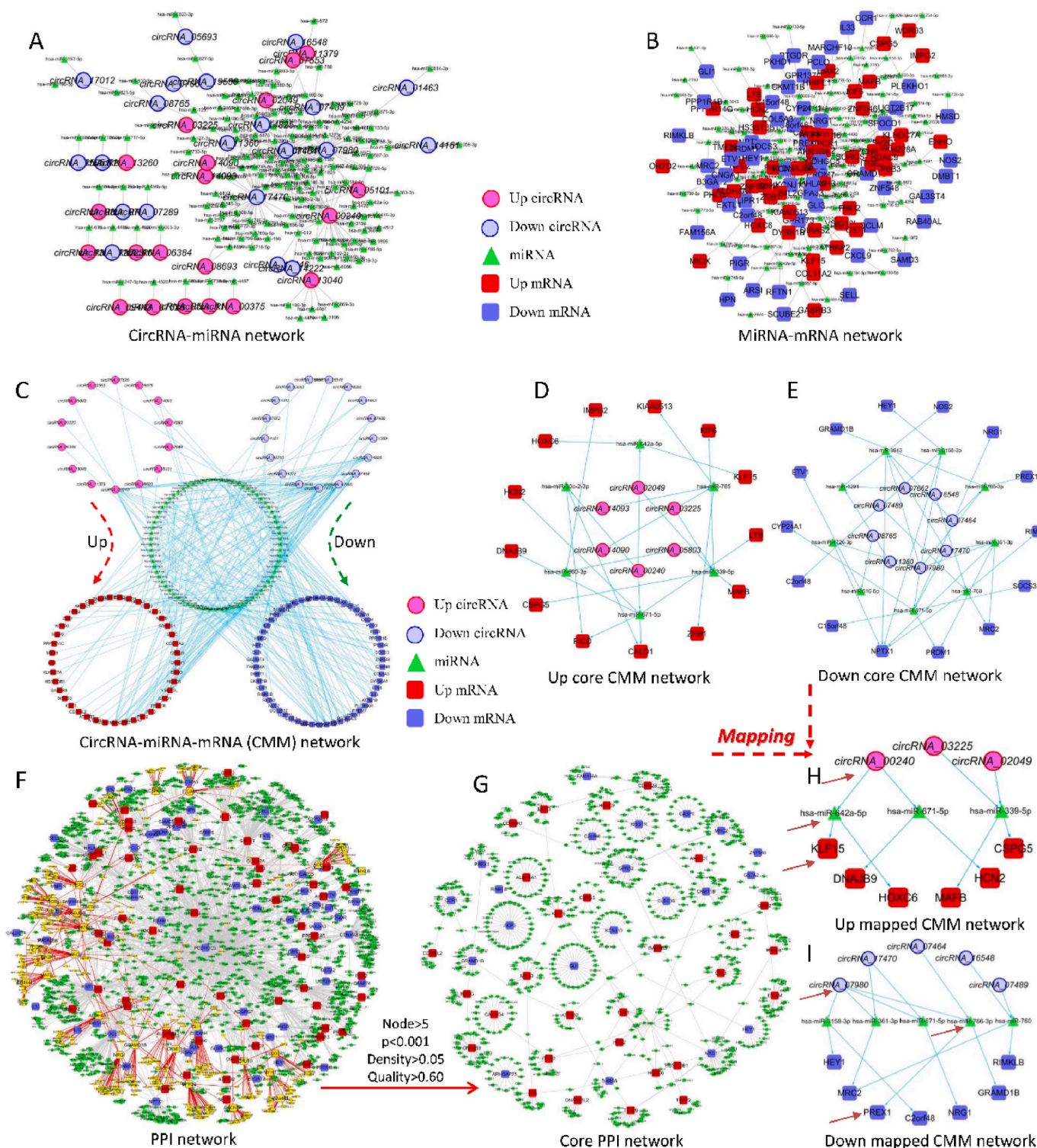


Fig. 2. The global ceRNA network was constructed based on the circRNA-miRNA-mRNA pairs and the core networks were rebuilt, respectively. (A) The global profiles of the HSYW formula treatment-related circRNA-miRNA network were built. (B) The global profiles of the HSYW formula treatment-related miRNA-mRNA networks were built. (C) The global circRNA-miRNA-mRNA (CMM) network was constructed; (D) Up core network was isolated from CMM network; (E) Down core network was isolated from CMM network; (F) The PPI network of HSYW treated group was constructed; (G) Core PPI network was isolated, and the selection criteria were defined as node > 5, $p < 0.001$, density > 0.05, quality > 0.60; (H) UP mapped CMM network of HSYW treated group; (I) Down mapped CMM network of HSYW treated group.

network (Fig. 3A and D). The results show that the circRNA_00240, circRNA_02049, KLF15, HCN2 and CSPG5 were upregulated in HSYW-treated N87-bearing Balb/c mice (Fig. 3B). Afterward, the down-regulated circRNA_07980, circRNA_07464, PREX1, C2orf48 and

GRAMD1B also were confirmed in HSYW-treated N87-bearing Balb/c mice (Fig. 3E). Using bioinformatics analysis, we found the up circRNA_00240 regulate to miR-642a-5p and miR-671-5p, and then targeting KLF15 and HCN2, respectively, and up circRNA_02049

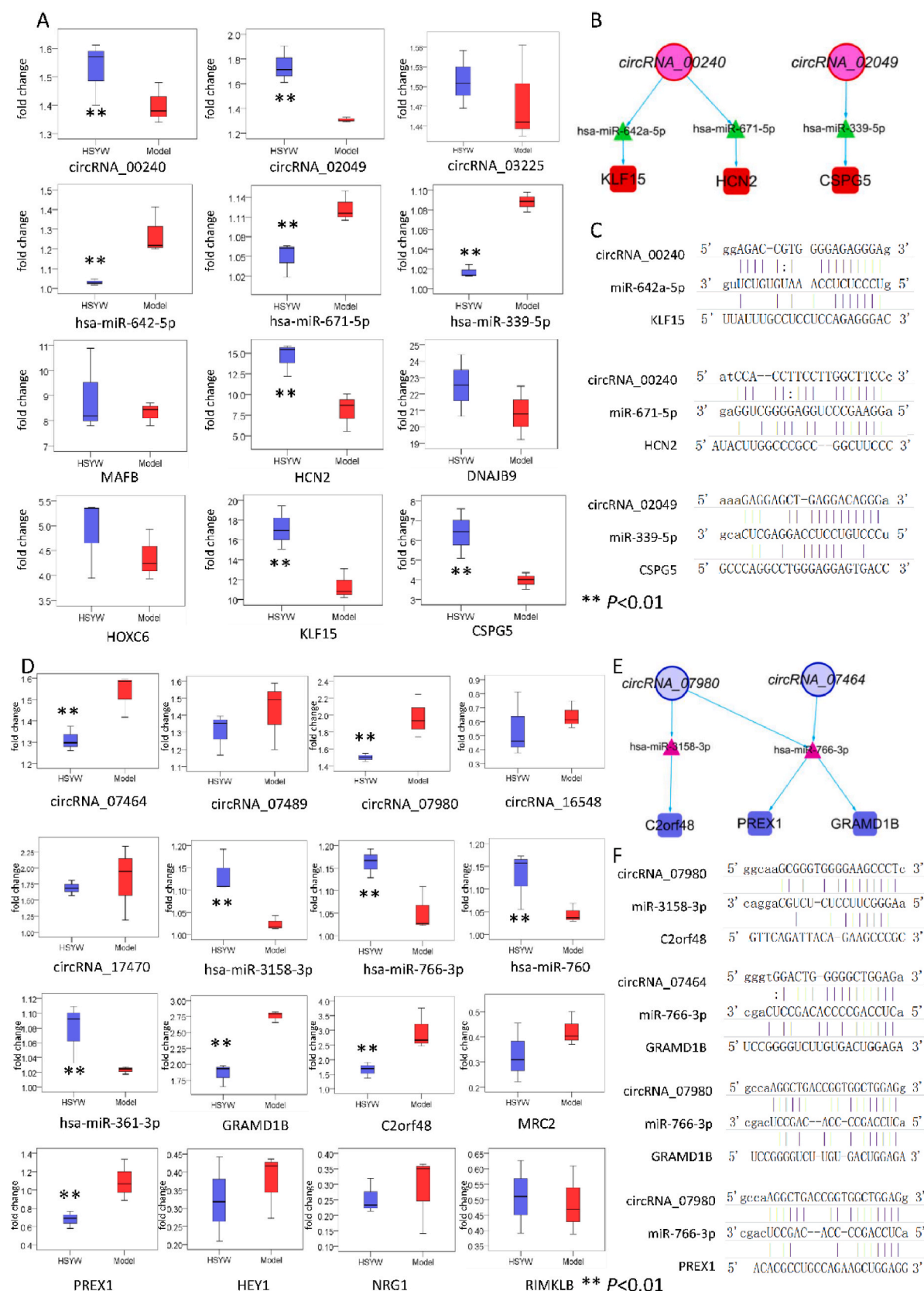


Fig. 3. The expression levels of circRNAs, miRNAs and mRNAs from core CMM network were validated by qRT-PCR. (A) The expression levels of upregulated circRNAs, miRNAs and mRNAs were verified by qRT-PCR; (B) The upregulated circRNA_00240 cluster and circRNA_02049 cluster was isolated from the core network; (C) The targeting relationship among the up circRNAs, miRNAs and mRNAs were predicted, respectively; (D) The expression levels of downregulated circRNAs, miRNA and mRNAs were verified by qRT-PCR; (E) The downregulated circRNA_020490 cluster and circRNA_07464 cluster was isolated from the core network; (F) The targeting relationship among the down circRNAs, miRNAs and mRNAs were predicted, respectively.

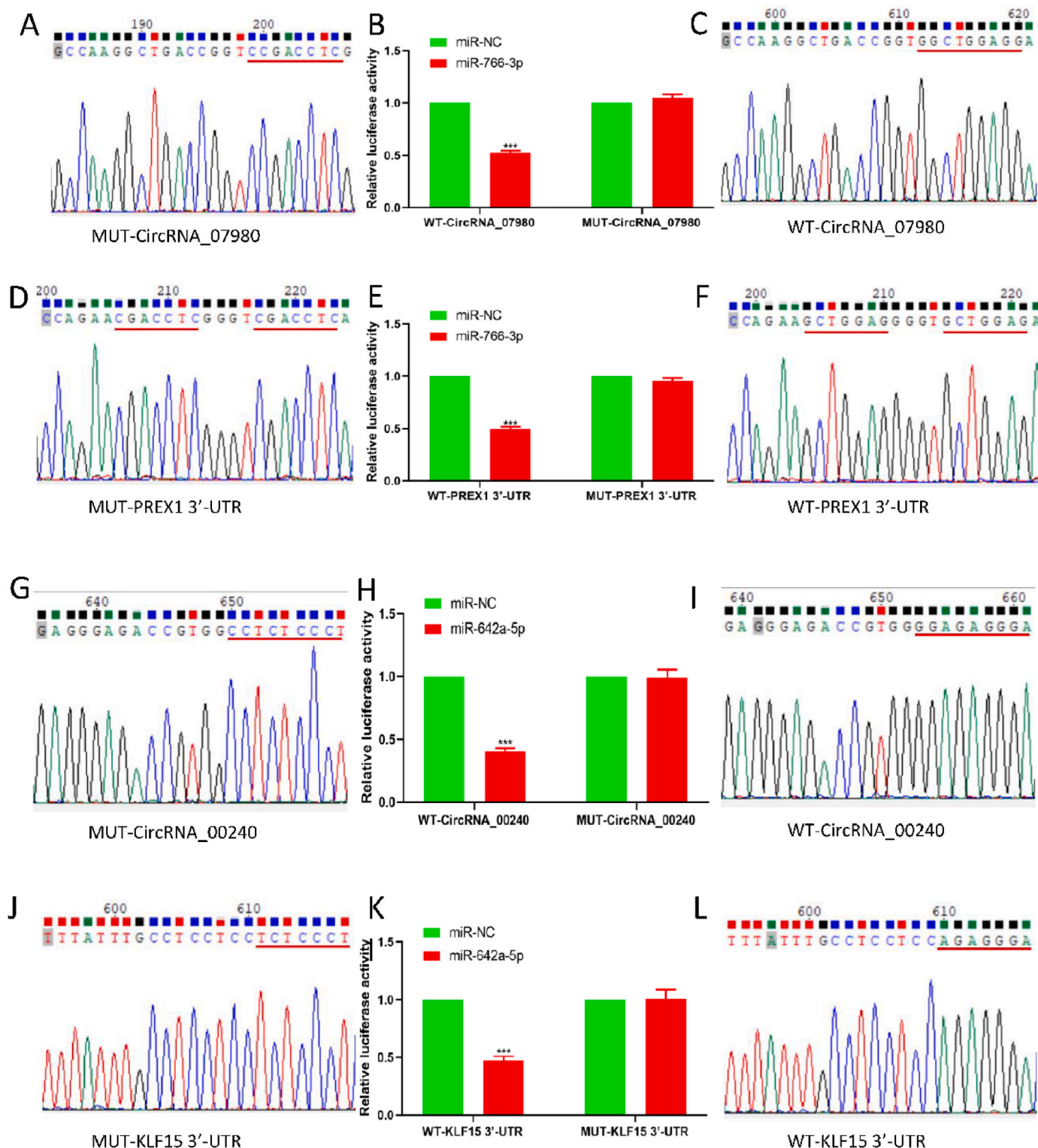


Fig. 4. The dual-luciferase reporter assay confirmed the connection of circRNA_00240/hsa-miR-642a-5p/KLF15 and circRNA_07980/hsa-miR-766-3p/PREX1 pathway respectively. A, C, D, F CircRNA_07980 or PREX1 wild type and mutant plasmids was constructed. B&F The link between circRNA_07980 and hsa-miR-766-3p/PREX1 was confirmed. G, I, J, K circRNA_00240 and KLF15 wild type and mutant plasmids was constructed. H&I The link between circRNA_00240, hsa-miR-642a-5p and KLF15.

regulate to miR-339-5p and targeting CSPG5 (Fig. 3C). As a contrast, the down circRNA_07980 and circRNA_07980 regulate to miR-3158-3p and miR-766-3p, and then targeting C2orf48, PREX1 and GRAMD1B, respectively (Fig. 3F).

3.7. Dual-luciferase reporter assay

To evaluate the binding ability of miRNAs to circRNA and mRNA, a luciferase assay was employed to validate the regulatory pathways involving circRNA_07980/hsa-miR-766-3p/PREX1 and circRNA_00240/

hsa-miR-642a-5p/KLF15. The results showed that the miR-766-3p significantly suppressed the luciferase activity of psiCHECK-2-WT-CircRNA_07980, while miR-NC had no inhibitory effect on psiCHECK-2-MUT-CircRNA_07980 (Fig. 4A–C), and the fluorescein activity of miR-766-3p and PREX1 has the same trend of difference (Fig. 4D–F). Meanwhile, the miR-642a-5p also significantly suppressed the luciferase activity of psiCHECK-2-WT-CircRNA_00240, while miR-NC had no inhibitory effect on psiCHECK-2-MUT-CircRNA_00240 (Fig. 4G–I), and the fluorescein activity of hsa-miR-642a-5p and KLF15 also has the same trend of difference (Fig. 4J–L). These results revealed that hsa-miR-766-3p can bind circRNA_07980 and PREX1, and hsa-miR-642a-5p can bind circRNA_00240 and KLF15, respectively.

3.8. Validation based on TCGA and HPA database

To further validate the status of these targeted mRNAs in GC, expression levels of 6 core genes between gastric cancer (n = 408) and normal (n = 36) were evaluated using TCGA and Human Protein Atlas (HPA) databases. The GEPIA online analysis revealed that the KLF15 and PREX1 had significantly differential expression levels between gastric cancer and normal (Fig. 5A). Although the expression levels of HCN2, CSPG5, C2orf48, and GRAMD1B have had little difference between tumor tissues and normal, the tumor grade analysis showed all targets had differential expressions among the three grades of gastric cancer (Fig. 5B). The tissue expression of these 6 core targets were then further evaluated via immunohistochemistry using the HPA database. The results showed that 5 targets were differential expressed between tumor and normal tissues, and HCN2 has not any information in HPA database. In the 5 targets, the KLF15 was negative in tumor tissues and strong expressed in normal, the CSPG5, C2orf48 and PREX1 were moderate in tumor and weak or negative in normal tissues, while the GRAMD1B was both moderate between tumor and normal tissues (Fig. 5C). Thus, combining these data, it can be concluded that the CSPG5, KLF15, C2orf48 and PREX1 are highly correlated to gastric carcinogenesis, and that the circRNA_00240/hsa-miR-642a-5p/KLF15 and circRNA_07980/hsa-miR-766-3p/PREX1 pathways might have critical regulated functions in HSYW-treated gastric cancer process (Fig. 5D).

4. Discussion

By combining transcriptomic and network analysis of HSYW-treated N87-bearing Balb/c mice, we mechanistically dissected specific-effects of herbal formula HSYW in inhibiting GC growth from a circRNA to mRNA level. We found that 119 circRNAs and 200 mRNAs have significant differentially expressed between HSYW-treated and model mice. Enrichment analysis revealed that pathways related to steroid biosynthesis, 25-hydroxycholecalciferol-24-hydroxylase activity, and regulation of mesenchymal cell proliferation might be more important for HSYW treated gastric cancer. 25-Hydroxycholecalciferol is the predominant circulating form of vitamin D, which can be converted into a potent steroid hormone, and has been linked to diseases outside of skeletal tissues such as cancer (Jeon and Shin, 2018; Reynolds et al., 2020; Milan et al., 2022). Furthermore, mesenchymal cells are known to be capable of undergoing epithelial-to-mesenchymal transition (EMT) under certain conditions, and are associated with tumor initiation, invasion, metastasis and drug resistance (Pastushenko and Blanpain, 2019). The evidences demonstrated that the miRNA-20b promoted EMT in GC partially by activating the Wnt pathway via targeting SUFU (Peng et al., 2021), and downregulation of circRNA_0023642 could decrease the EMT-related gene in GC (Zhou et al., 2018). Overall, These findings point to 25-Hydroxycholecalciferol and EMT regulation may be important factors for HSYW inhibited gastric tumor growth *in vivo*.

We analyzed the topological structures of CMM network and PPI network, and 4 circRNAs, 5 miRNAs and 6 mRNAs were confirmed play critical roles in HSYW-treated N87-bearing Balb/c mice by qRT-PCR

validation. With targeting relationship analysis, we identified 7 circRNA-miRNA-mRNA pathways were high correlated to HSYW-treated gastric cancer. Among these pathways, upregulated circRNA_00240 was found to regulate miR-642a-5p and miR-671-5p, which in turn target KLF15 and HCN2 respectively. Similarly, upregulated circRNA_02049 was found to regulate miR-339-5p, which targets CSPG5. Additionally, downregulated circRNA_07980 and circRNA_07464 were observed to regulate miR-3158-3p and miR-766-3p, and subsequently target C2orf48, PREX1, and GRAMD1B, respectively. By examining the expression levels of the targeted mRNAs in the TCGA database, we found a significant difference in the expression levels between GC patients and healthy individuals for KLF15 and PREX1. Furthermore, the tumor grades analysis of the dataset revealed that all targets had differential expression levels between the three grades of gastric cancer.

We further analyzed the expression levels of these targeting mRNAs in TCGA database, and the results reveal that the KLF15 and PREX1 have significantly difference between GC patients and normal. The tumor grades analysis also show that all targets have differential expressions among the three grades of gastric cancer. Based on HPA database, we also found the KLF15 was negative in tumor tissues and strong expressed in normal, while the PREX1 were moderate in tumor and weak in normal tissues. Importantly, we found that the TCGA and HPA analysis results were high corresponded to our qRT-PCR validations. This phenomenon suggests that circRNA_00240/hsa-miR-642a-5p/KLF15 and circRNA_07980/hsa-miR-766-3p/PREX1 are two essential regulatory axes in HSYW-treated gastric cancer processes. The dual-luciferase assay also showed that the miR-766-3p can bind to circRNA_07980/PREX1, and miR-642a-5p can bind to circRNA_00240/KLF15.

In the circRNA_00240/hsa-miR-642a-5p/KLF15 axis, the KLF15 (Krueppel-like factor 15) plays a significant role in cancer progression, and over-expression of KLF15 could inhibit GC cell proliferation partly via regulating CDKN1A/p21 and CDKN1C/p57 (Sun et al., 2017). Another important axis is that PREX1 belongs to the P-Rex family of Rho guanine nucleotide exchange factors (Rho GEFs). Its activity is synergistically regulated by phosphatidylinositol (3,4,5)-trisphosphate (PIP3) and Gβγ (Lucato et al., 2015; Barrows et al., 2016). PREX1 has also been shown to promote tumor development, invasion and metastasis, particularly in breast cancer and melanoma (Noack et al., 2012; Clements and Johnson, 2020). Moreover, PREX1 had a significant positive correlation with clinical diagnostic and prognostic evaluation in GC (He and Wang, 2020; Li et al., 2021; Liu et al., 2020). with evidence pointing to the stimulation of tumor migration and invasive activity via the activation of the Transforming Growth Factor Beta 1 (TGFβ1) pathway (Shao and Chen, 2020).

In summary, our study demonstrated that HSYW significantly inhibited tumor growth *in vivo*. Transcriptional profiling and systems network show that 4 circRNAs, 5 miRNAs and 6 mRNAs act as potential markers to evaluate the curative effects of HSYW against tumor growth in N87-bearing Balb/c mice. By combining the experimental and bioinformatics analysis, we confirm that the circRNA_00240/hsa-miR-642a-5p/KLF15 and circRNA_07980/hsa-miR-766-3p/PREX1 pathways play critical roles in HSYW-treated gastric cancer. However, it is crucial to acknowledge that further exploration is necessary to screen specific monomers from HSYW for further in-depth analysis.

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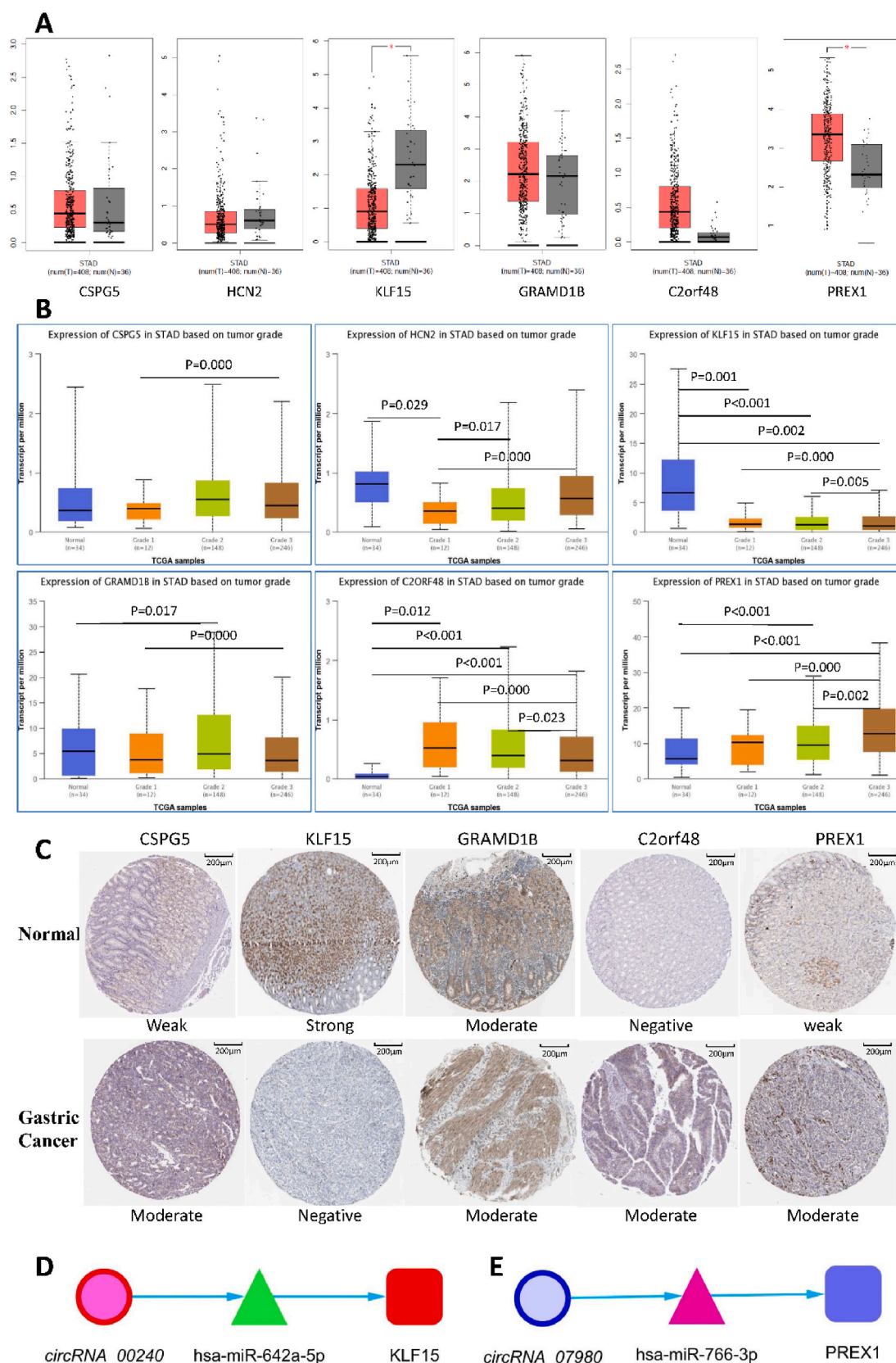


Fig. 5. The core mRNAs were validated based on TCGA and HPA databases. (A) The expression of KLF15, HCN2, CSPG5, PREX1, C2orf48, and GRAMD1B were calculated based on GEPIA database (Tumor group, $n = 408$; Normal group, $n = 36$); (B) Expression analysis for KLF15, HCN2, CSPG5, PREX1, C2orf48, and GRAMD1B among different tumor grades were calculated in gastric cancer based on UALCAN databases. (C) The immunohistochemistry results of the KLF15, CSPG5, PREX1, C2orf48, and GRAMD1B between human gastric glandular cells and gastric cancer cells; (D) The tendency of circRNA_00240/hsa-miR-642a-5p/KLF15 pathway in HSYW-treated N87-bearing Balb/c mice; (E) The tendency of circRNA_07980/hsa-miR-766-3p/PREX1 pathway in HSYW-treated N87-bearing Balb/c mice.

CRediT authorship contribution statement

Mengyuan Zhang: Methodology, Writing – original draft. **Yujie Ding:** Resources, Writing – original draft. **Sheng Hu:** Data curation, Visualization. **Fulong Li:** Formal analysis, Software. **Yi Wang:** Supervision, Validation. **Yue Zhou:** Supervision. **Mei Qi:** Validation. **Hongmei Ni:** Methodology, Supervision. **Shengquan Fang:** Methodology, Funding acquisition, Resources. **Qilong Chen:** Conceptualization, Methodology, Writing- Reviewing and Editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

Data availability

Data will be made available on request.

Abbreviations

ceRNAs	Competing for endogenous RNAs
circRNAs	Circular RNAs
CMM	circRNA-miRNA-mRNA
CYPs	Cytochrome P450 enzymes
DE	Differentially expressed
EMT	Epithelial-mesenchymal transition
FBS	Fetal bovine serum
FPKM	Fragments Per kb Per Million Reads
GBD	Global Burden of Disease
GC	Gastric cancer
GEFs	Guanine nucleotide exchange factors
GEPIA	Gene Expression Profiling Interactive Analysis
GTEX	Genotype-Tissue Expression
GO	Gene Ontology
HPA	Human Protein Atlas
HPLC	High Performance Liquid Chromatography
HSYW	Huosu-Yangwei
HSYW-L	low dose HSYW
HSYW-M	medium dose HSYW
HSYW-H	high dose HSYW
HPA	The Human Protein Atlas
KEGG	Kyoto Encyclopedia of Genes and Genomes
MF	Molecule function
miRNA	MicroRNA
NB	Negative binomial distribution
OXA	Oxaliplatin
PPI	Protein-protein interaction
qRT-PCR	Quantitative real-time PCR
RNA-seq	RNA sequencing
RPM	Spliced reads per million
SOX S-1	plus oxaliplatin
TCGA	The Cancer Genome Atlas
TCM	Traditional Chinese medicine

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2023.116674>.

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